

STAT 5700 — Quiz 7**Date:** April 13, 2026**SOLUTIONS****FOR ALL 3 PROBLEMS:** Let X and Y be two continuous random variables with joint pdf

$$f(x, y) = 3xy, \quad 0 < x < y, \quad x + y \leq 2$$

Problem 1

We are given independent random variables with densities

$$f_X(x) = 2x, \quad 0 < x < 1 \quad \text{and} \quad f_Y(y) = 3y^2, \quad 0 < y < 1.$$

We want:

$$\begin{aligned} P(X < Y) &= \int_0^1 \int_0^y f_X(x) f_Y(y) \, dx \, dy \\ &= \int_0^1 \int_0^y (2x)(3y^2) \, dx \, dy = \int_0^1 3y^2 \left(\int_0^y 2x \, dx \right) dy \\ &\quad \int_0^y 2x \, dx = y^2 \end{aligned}$$

So:

$$P(X < Y) = \int_0^1 3y^2 \cdot y^2 \, dy = \int_0^1 3y^4 \, dy = 3 \cdot \frac{1}{5} = \frac{3}{5}$$

Answer: $\frac{3}{5}$ **Problem 2**

Given:

$$\mu_X = 2, \quad \sigma_X^2 = 9, \quad \mu_Y = 1, \quad \sigma_Y^2 = 16, \quad \sigma_{XY} = 3$$

(a)

$$\text{Cov}(X, 2Y) = 2 \cdot \text{Cov}(X, Y) = 2 \cdot 3 = 6$$

(b)

$$\text{Corr}(X, 2Y) = \frac{\text{Cov}(X, 2Y)}{\sigma_X \sigma_{2Y}}$$

Note:

$$\sigma_X = 3, \quad \sigma_{2Y} = 2\sigma_Y = 2 \cdot 4 = 8$$

So:

$$\text{Corr}(X, 2Y) = \frac{6}{3 \cdot 8} = \frac{6}{24} = \frac{1}{4}$$

Answers: (a) 6, (b) $\frac{1}{4}$

Problem 3

Which can be negative?

- μ_X (mean): **can be negative**
- σ_X (standard deviation): **cannot be negative**
- $\text{Cov}(X, Y)$: **can be negative**
- ρ_{XY} (correlation): **can be negative**

Answer: A, C, D

Problem 4

If correlation is exactly 1, then the relationship is:

$$Z = aW + b \quad \text{with } a > 0$$

So the slope must be **positive**.

- Could slope be 0.5? **Yes**
- Could slope be -2 ? **No**

Answer: B